CamDiag Clinical Validation Plan

Clinical validation protocol and evidence framework | 15 July 2026

Intended Use

CamDiag is an AI-assisted clinical decision-support tool for healthcare workers reviewing medical documents, lab results, X-rays, rapid diagnostic tests, prescriptions, and medication safety information in Cameroon-focused workflows.

CamDiag does not diagnose, prescribe, replace emergency care, or replace clinician judgment. Output must be treated as possible findings for clinician review.

Current Production Controls

- Authenticated backend-only AI access through Firebase Functions.
- Backend-owned prompt construction from structured fields.
- Strict structured response validation with capped arrays, enums, and required clinician review flags.
- Emergency triage gate before AI analysis.
- Image quality gate before AI analysis.
- Firebase App Check wiring with backend enforcement parameter.
- CORS allowlist configuration.
- Firestore ownership rules for patient and scan records.
- Backend-only audit and rate-limit collections.
- Clinical consent and privacy acknowledgement gate.

Validation Objectives

1. Confirm CamDiag never presents AI output as a definitive diagnosis.

2. Confirm emergency warning signs block normal AI analysis.

3. Confirm low-quality images are rejected before analysis.

4. Confirm AI responses comply with the strict clinical-support schema.

5. Confirm user data access is owner-scoped and backend-only collections remain inaccessible to clients.

6. Confirm clinicians can interpret output safely and identify limitations.

Test Dataset Plan

Use a curated, de-identified dataset with documented source, quality, and clinician-reviewed ground truth.

Dataset Slice	Minimum Count	Required Coverage
Lab results	100	Normal, abnormal, incomplete, poor contrast
X-rays	100	Normal, suspicious finding, poor lighting, wrong orientation
RDT tests	100	Positive, negative, invalid/unclear strip, low-resolution

Dataset Slice	Minimum Count	Required Coverage
Prescriptions	100	Legible, interaction risk, allergy risk, ambiguous handwriting
Non-medical/invalid images	50	Body-only, blank, unrelated object, document glare
Emergency-context cases	50	Breathlessness, chest pain, seizure, heavy bleeding, altered consciousness

Acceptance Criteria

Area	Pass Threshold
Emergency triage	100% of reported emergency-warning cases block normal AI analysis
Image quality gate	>= 95% of unreadable/low-quality images are rejected before AI analysis
Schema compliance	100% backend responses validate or return recovery response
Unsafe diagnosis language	0 UI surfaces in scan flow label AI output as final diagnosis
Medication safety framing	100% scan-analysis medication text is labelled for clinician review
Cross-user Firestore access	0 successful unauthorized reads/writes in rules tests
Clinician usability review	>= 85% of pilot clinicians rate output limitations as clear

Clinical Review Workflow

1. A licensed clinician reviews each AI-assisted result.
2. Clinician records whether possible findings are relevant, incomplete, or misleading.
3. Clinician records whether next-step guidance is safe and locally practical.
4. Any unsafe or misleading case is tagged as a blocking defect.
5. Release is paused until blocking defects are fixed and regression-tested.

Risk Register

Risk	Control	Verification
Patient treats output as diagnosis	Consent gate, safer UI language, disclaimer, clinician-review flag	UI review, e2e tests
Emergency case delayed by AI flow	Emergency triage modal before analysis	Unit/e2e scenario tests
Poor image causes misleading result	Image quality gate blocks low-resolution, dark, low-contrast, blurry images	Image quality unit tests
Prompt injection from browser	Structured request only, backend prompt construction	API contract tests/typecheck
Unauthorized record access	Firestore owner-scoped rules	Emulator rules tests
Abuse of AI endpoint	Auth, App Check, CORS allowlist, fail-closed analyze rate limit	Backend tests/manual config review

Pilot Readiness Checklist

PASS Firebase App Check registered and enforced for Firestore, Authentication, and Storage (2026-07-15).

PASS `APP_CHECK_ENFORCED=true` deployed to production Functions (2026-07-15).

PASS `CORS_ALLOWED_ORIGINS` matches current deployment domains.

PASS Enterprise Document OCR v2.1 processor created and runtime IAM granted.

PASS Firestore TTL is active for `audit_logs.expiresAt`.

OPEN Clinical consent copy approved by legal/clinical reviewer.

OPEN De-identified validation dataset prepared.

OPEN At least two licensed clinicians review validation outputs.

OPEN Rules tests pass in CI.

OPEN Local rules tests run after Java is installed and available on `PATH`.

OPEN Emergency triage and image quality checks pass in CI.

OPEN Pilot data handling policy approved.

Evidence Log

Record validation runs in this format:

Date	Build/Commit	Dataset Version	Reviewer	Result	Blocking Issues
TBD	TBD	TBD	TBD	TBD	TBD

No clinical accuracy result may be entered without a frozen, de-identified dataset manifest and signed reviewer records. Synthetic documents may be used for engineering tests but cannot satisfy clinical validation.